

ASCEND INSTITUTE, GANGTOK

CHAPTER-WISE TEST

ICSE CLASS 8 - MATHEMATICS

Chapter: Identities

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. The expansion of $(x + 4)^2$ is:

(i) $x^2 + 8x + 16$

(k) $x^2 + 16$

(j) $x^2 + 4x + 16$

(l) $x^2 - 8x + 16$

Answer: _____

2. $(3a - 2b)^2$ is equal to:

(i) $9a^2 - 12ab + 4b^2$

(k) $9a^2 + 12ab + 4b^2$

(j) $9a^2 - 6ab + 4b^2$

(l) $6a^2 - 12ab + 4b^2$

Answer: _____

3. Complete: $(2p + q)^2 = 4p^2 + \text{_____} + q^2$.

(i) $2pq$

(k) $6pq$

(j) $4pq$

(l) $8pq$

Answer: _____

4. Which expansion is incorrect?

(i) $(x + 3)^2 = x^2 + 6x + 9$

(k) $(2x + 1)^2 = 4x^2 + 2x + 1$

(j) $(x - 5)^2 = x^2 - 10x + 25$

(l) $(a - b)(a + b) = a^2 - b^2$

Answer: _____

5. $x^2 - 49$ can be written as:

(i) $(x - 7)^2$

(k) $(x - 7)(x + 7)$

(j) $(x + 7)^2$

(l) $(x - 49)(x + 1)$

Answer: _____

6. Which expression equals $(p + q)^2 - 2pq$?

(i) $p^2 + q^2$

(k) $2pq$

(j) $p^2 - q^2$

(l) $p + q$

Answer: _____

7. If $a + b = 9$ and $ab = 14$, then $a^2 + b^2 =$

(i) 39

(k) 67

(j) 53

(l) 81

Answer: _____

8. If $x - y = 6$ and $xy = 5$, then $x^2 + y^2 =$

(i) 26

(k) 46

(j) 36

(l) 56

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Expand and simplify: $(4x + 3y)^2$. [2]

10. Expand and simplify: $(5a - 2b)^2$. [2]

11. Simplify using identities: $(2x + 3)^2 - (2x - 3)^2$. [2]

12. Factorise completely: $81p^2 - 16q^2$. [2]

13. If $a + b = 11$ and $ab = 24$, find $a^2 + b^2$. [2]

14. A square has side $(2m - 3)$ cm. Find its area in simplified form. [2]
